NOTICE OF CHANGE

MIL-HDBK-1028/1
NOTICE 1
20 March 1990

MILITARY HANDBOOK

AIRCRAFT MAINTENANCE FACILITIES

TO ALL HOLDERS OF MIL-HDBK-1028/1:

1. THE FOLLOWING PAGES OF MIL-HDBK-1028/1 HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
13	20 March 1990	13	REPRINTED WITHOUT CHANGE
14	20 March 1990	14	15 November 1989
14a	20 March 1990	new page	
14b	20 March 1990	new page	
15	20 March 1990	15	15 November 1989
16	20 March 1990	16	15 November 1989
17	20 March 1990	17	15 November 1989
17a	20 March 1990	new page	
17b	20 March 1990	new page	
18	20 March 1990	18	15 November 1989
18a	20 March 1990	new page	
18b		new page	
57	20 March 1990	57	15 November 1989
58	20 March 1990	58	REPRINTED WITHOUT CHANGE
72a	20 March 1990	new page	
72b	20 March 1990	new page	
72c	20 March 1990	new page	
72d	20 March 1990	new page	
72e	20 March 1990	new page	
72f	20 March 1990	new page	
72g	20 March 1990	new page	
72h	20 March 1990	new page	
153	20 March 1990	153	15 November 1989
153a	20 March 1990	new page	
153b	20 March 1990	new page	
154	20 March 1990	154	REPRINTED WITHOUT CHANGE

2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of MIL-HDBK-1028/1 will verify that all changes indicated above have been made. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication.

Source: https://assist.dla.mil -- Downloaded: 202640-26T13:56Z
Check the source to verify that this is the currentversion before use.

Each notice is to be retained by stocking points until the military handbook is completely revised or cancelled.

CUSTODIAN:
NAVY-YD

PREPARING ACTIVITY:
NAVY-YD

PROJECT NO.
FACR-0374

AMSC N/A

AREA FACR

DISTRIBUTION STATEMENT A. APPROVED FOR PUBLIC RELEASE: DISTRIBUTION UNLIMITED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

2.7 Security. The Maintenance Hangar shall contain secure storage vaults with walls of 8-in. (203 mm) masonry extending to the roof level. Vaults shall meet security requirements of the facility in accordance with Chapter V of OPNAVINST 5510.1G, Department of the Navy Information Security Program Regulation. In Marine Corps hangars, secure space shall be provided for organizational small arms.

2.8 Weight-Handling Equipment. The Maintenance Hangar shall contain overhead bridge cranes and jib cranes in accordance with criteria in NAVFAC DM-38.01, Weight-Handling Equipment, and paras. 2.8.1 and 2.8.2 for Types I and II (OH) spaces.

2.8.1 Bridge Cranes. Overhead bridge cranes shall have an electric motorized bridge, trolley, and hoist and shall be 6-ton (5,443 kg) capacity. Trolley and bridge shall be capable of operating at a slow speed of 15 to 20 ft/min (0.076 to 0.102 m/s) for positioning loads and at a high speed for moving loads of 60 ft/min (0.305 m/s); the hoist shall be capable of operating at a slow speed of 3 to 4 ft/min (0.015 to 0.02 m/s) and at a high speed of 12 ft/min (0.06 m/s). Refer to NAVFAC NFGS-14637, Cranes, Overhead Electric, Underrunning (Under 20,000 Pounds).

2.8.1.1 Motor and Controls. Bridge, trolley, and hoist controls shall provide for two-speed reversing of a two-speed, squirrel-cage motor. The controls shall be equipped with reduced voltage starting for the motors. Controls shall be operable from the floor.

2.8.1.2 Hook Height. The hook height for bridge cranes shall be as required to raise engines and transmissions clear of aircraft.

2.8.2 Jib Cranes. Jib cranes shall be electrically operated with a capacity of 500 pounds (226.5 kg). Jib cranes shall be capable of transferring a palletized load from the hangar deck to the mezzanine level.

PRINTED WITHOUT CHANGE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

Section 3: CORROSION CONTROL HANGAR

3.1 Function. The Corrosion Control Hangar shall be designed to provide space and equipment for the corrosion control processing of aircraft at the intermediate maintenance level. Functions performed in the Corrosion Control Hangar include: deicing, limited detergent washing and rinsing, paint stripping, corrosion removal, protective coating application and painting, and finish curing and drying. Refer to NAVAIR Technical Manual 01-1A-509, Aircraft Weapons Systems Cleaning and Corrosion Control, for detailed functions performed in this facility.

3.2 Location. The Corrosion Control Hangar should be located in proximity to the maintenance hangars and as close as possible to an aircraft washrack. Access between the Corrosion Control Hangar, the Maintenance Hangar, and the aircraft washrack is required. The prevailing wind, other buildings, and site geological features should be considered in orienting the building in relation to aprons, taxiways, and parking, to allow for exhaust air dispersal over areas not affected by solvent fumes.

3.3 Architectural Requirements. Two types of corrosion control hangars are indicated on the Facility Plates (Appendix A). The Type A Corrosion Control Hangar is designed for carrier aircraft and other small aircraft, and the Type B Corrosion Control Hangar is designed for land-based patrol aircraft. Because both construction and operating costs increase as the volume of the hangar bay increases, corrosion control hangars for aircraft larger than those accommodated by the Type B hangar shall be individually designed around the specific dimensions of the aircraft concerned. The general architectural criteria of Section 2, covering maintenance hangars, are applicable to the Corrosion Control Hangar and shall be followed.

a) The size of various aircraft scheduled to use the facility will determine the hangar bay dimensions. Provide the following minimum clearances:

(1) Wingtip/rotor blade to wall each side - 10 feet (3.05 m)

(2) Top of aircraft (vertical fin, radome, rotor head, tail rotor) to ceiling - 6 feet (1.83 m)

(3) Nose of aircraft to hangar door - 10 feet (3.05 m)

(4) Tail of aircraft or tail rotor to exhaust target wall - 10 feet (3.05 m)

These clearances are minimal to allow proper access for work platforms and to minimize paint overspray on hangar walls and ceilings. Aircraft dimensions shall be based on the maximum condition (wings/rotors spread versus folded) unless it has been determined that aircraft surfaces are accessible with the wings/rotors folded.

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the current version before use.

b) The number of hangar bays for each site shall be based on an analysis of aircraft types, production schedules, hours required for each corrosion control operation and number of work shifts.

c) Ancillary space requirements will vary based on facility requirements. Spaces such as parts stripping, parts painting, and non-destructive testing may be required. These spaces shall be provided as required and design shall conform to that for similar spaces described herein. Provide exit doors to the outdoors for rooms designated for storing or mixing chemicals or paints. Provide a depressed floor or door sills with ramps to contain spills.

d) Provide storage space for dry filters. Consider a service elevator if rooftop heat/ventilation units are used to provide for personnel and materials transport to roof. Otherwise provide stairs for personnel and a roof mounted jib crane for materials transport to roof.

e) Consider providing a loading dock.

f) Comply with environmental protection regulations.

3.3.1 Structure. The hangar structure shall be steel frame with open-web, steel-joist roof. The structure for areas adjacent to the hangar area shall be load-bearing, insulated, concrete-masonry unit construction with open-web, steel-joist roof. The strip/rinse room shall be equipped with a 2-ton (1,814-kg) overhead monorail. The requirements of NFPA-33, Spray Application Using Flammable and Combustible Materials, shall be incorporated into the design.

3.3.2 Walls. Hangar walls shall meet the following criteria:

a) Exterior walls of the hangar area shall be a steel frame and girts framing system with insulated preformed (corrugated) metal siding. Exterior walls of the other areas shall be insulated concrete masonry units.

b) The hangar area shall be separated from paint mixing and storage rooms, chemical mixing rooms, cleaning gear and storage room, corridor, and toilet and locker areas, by 2-hour, fire-rated walls.

c) Hangar bays in multiple bay structures shall be separated by 3-hour, fire-rated walls.

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

3.3.3 Roofs. Roofs shall meet the criteria defined in para. 2.3.2.

3.3.4 Floors. The hangar area floor shall be designed in accordance with criteria in NAVFAC DM-21 Series and may have a reflective floor coating. Other floors shall be designed in accordance with the NAVFAC DM-2 Series, Structural Engineering, and provided with floor coverings as specified in MIL-HDBK-1001/2.

3.3.5 Ceilings. All aircraft bays and paint storage, mixing, and spray areas shall have water-resistant, airtight gypsum ceilings. Suspended acoustical ceilings shall be provided for corridors, toilets, locker rooms, and offices. Suspended acoustical ceiling panels in shower areas shall have vinyl plastic surfaces. The strip/rinse room ceiling shall be of corrosion-resistant, perforated metal.

3.3.6 Doors and Door Controls. Doors and controls shall be as described in paras. 3.3.6.1 through 3.3.6.3.

3.3.6.1 Hangar Doors. Hangar doors shall be a specialized, insulated, plenum type with swinging leaves. These doors shall serve as an insulated supply air plenum when closed. Each door leaf shall be a motor-operated unit with a release mechanism, and the doors shall be provided with a means of movement in the event of a power failure. Adequate safety devices shall be installed to prevent injury to personnel and damage to equipment due to moving door sections. Thresholds shall be designed to minimize dirt accumulation and ice buildup.

3.3.6.2 Personnel Doors. Personnel doors between the hangar area and paint and chemical mixing rooms and the corridor shall be 1-1/2-hour, B-rated, self-closing fire doors. Hold-open devices shall not be permitted.

3.3.6.3 Other Doors. Other doors shall be as described in para. 2.3.6.5. Provide self-closing, insulated personnel exit doors from each hangar bay. No hold-open devices shall be permitted. Special precautions shall be taken to seal doors between hangar areas and exterior or adjacent spaces. A vehicle access door shall be provided either within or adjacent to the hangar aircraft access doors.

3.3.7 Building Insulation. Building walls, roofs, and floors shall meet the criteria defined in para. 1.6.

3.3.8 Surface Treatment. In addition to the requirements of para. 2.3.8, the side walls, the inside of hangar doors, and the ceiling of the hangar area shall be provided with a light color, smooth surface such as white enameled metal panels.

3.3.9 Exterior Pavement. Exterior paved areas include aircraft and vehicle access and nonorganizational parking. Aircraft pavement shall be in accordance with the NAVFAC DM-21 Series. Vehicle access and parking shall be in accordance with NAVFAC P-272, Parts 1 and 2; NAVFAC DM-5.04; and MIL-HDBK-1190.

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the currentversion before use.

3.4 Mechanical Requirements. The design of the Corrosion Control Hangar shall meet the mechanical requirements set forth in paras. 3.4.1 through 3.4.5.

3.4.1 Heating. Heating shall be provided in accordance with NAVFAC DM-3.03 and as follows:

a) Based on an economic analysis, consideration shall be given to providing a system for the recovery of heat from the exhaust air.

b) The required hangar bay design heating temperature will normally be between 70° F (24° C) and 90° F (32° C). Some activities may require higher temperatures to accelerate curing cycles. Exact requirements for each site must be determined based on materials and production requirements. The hangar area shall be equipped with an automatic cutoff of the heating system when the main hangar doors are open.

Minimum temperature thermostat shall be set at 34° F (1° C) to override the heating deactivation switch during door-open periods of subfreezing ambient temperatures. The recovery time for the heating system shall be 30 minutes after door closure.

c) Provide a snow-melting system at the hangar door tracks when the outside design temperature is plus 25° F (-4° C) or lower and when historical snow accumulation data supports the requirement.

d) Refer to Facility Plates for design conditions in other hangar spaces.

e) Each bay in the OH area shall be a separate heating zone.

f) Heating systems shall be suitable for operation in the vapor hazard condition in the hangar bay, flammable storage/mixing areas, and paint equipment cleaning spaces.

3.4.2 Ventilation for Control of Air Contaminants and Flammable Vapors. Ventilation shall be provided in accordance with Industrial Ventilation Systems, NAVFAC MIL-HDBK-1003/17 and NFPA-33. The ventilation system for the hangar area shall furnish 100 percent filtered outside air at a horizontal laminar flow velocity of 100 FPM (0.51 m/s) across the entire cross-section area of the stripping/painting hangar bays for worker safety and paint overspray control. Provide 50 FPM (0.25 m/s) airflow velocity during the drying cycle and a minimum of 6 air changes per hour at other times. The ventilation system shall maintain a slight negative static pressure of 0.05 in. (1.25 mm) water gage between the hangar area and exterior or adjacent spaces. Ventilate the space above the hangar bay ceiling to provide a nonhazardous space for light fixtures.

Filters throughout shall be made from noncombustible materials and shall meet approval of fire protection authorities. Supply air filters having a minimum atmospheric dust spot efficiency of 30 percent on the basis

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

of ASHRAE Standard 52-76 shall be provided at the intake to air handling units and in door plenum discharge. The type of exhaust filter system shall be determined by an analysis of environmental control regulations at the site. The use of air-assisted airless or electrostatic paint spray methods versus conventional air atomization is recommended to reduce overspray and increase exhaust filter life. The disposal and replacement cost of the replaceable dry-type filters shall be compared with the sludge disposal and water treatment costs of a water wash filter system. Where dry-type exhaust filters are used, visual gauges, audible alarms, or pressure activated devices shall be installed to ensure that the air velocity is maintained. Use of a water wash system will prevent effective heat recovery. Exhaust ventilation from floor level shall be provided for the paint mixing and storage rooms at the rate of 1 cfm (0.00047 m^3/s) per square foot of floor area or 150 cfm (0.071 m^3/s), whichever is greater. The ventilation systems for paint mixing and paint equipment cleaning rooms shall be provided in accordance with NFPA 33 and the American Conference of Governmental Industrial Hygienists' (ACGIH), Industrial Ventilation, A Manual of Recommended Practice. Consider heat recovery in the ventilation system for those spaces. Exhaust stacks shall be the "no loss" type as shown in ACGIH, Industrial Ventilation, A Manual of Recommended Practice.

3.4.3 Plumbing. Plumbing shall be provided in accordance with NAVFAC DM-3.01 and as follows:

a) Provide steam and hot and cold water.

b) Provide an emergency eyewash/safety shower conforming to ANSI Z358.1, American National Standard for Emergency Eyewash and Shower Equipment, and floor drains in hangar bays, paint mixing rooms, and paint equipment cleaning rooms.

c) Provide an oil separator for drains from the hangar area and paint mixing rooms and paint equipment cleaning rooms.

d) Storm-water drains should be located a minimum of 12 in. (305 mm) from the hangar access door rails.

3.4.4 Compressed Air. Provide in accordance with NAVFAC DM-3.05.

a) Provide low pressure compressed air at 125 psig (862 kPa) for shop use. Air shall be oil-free to prevent paint contamination. Air outlets supplying tools requiring lubrication shall be equipped with an in-line lubricator. Rotary oil-free compressors are recommended since this compressed air may be used as a source for breathing air at lower life cycle costs.

b) Provide low pressure compressed air at 20 psig (138 kPa) or higher if required, for breathing air in the hangar bays. Breathing air may be obtained from the oil-free shop air source through final purifiers in each bay or from a separate breathing air compressor and piping system. Breathing air from the oil-free shop air source is preferred because of lower cost.

Source: https://assist.dla.mil -- Downloaded: 202640-26T13:56Z
Check the source to verify that this is the current version before use.

c) The air outlets (quick connect fittings) for oil-free shop air, lubricated tool air and breathing air shall be different for each service and shall not be compatible with each other.

d) Typical outlet quantities for each bay are: four breathing and four oil-free shop air and two lubricated tool air. Typical outlet quantities for ancillary spaces requiring air are two oil-free shop air. Verify actual requirements for each site.

e) Ensure the intake for breathing air is located in an uncontaminated area.

3.4.5 Noise and Vibration Control. Mechanical systems and equipment shall be designed to limit noise and vibration in accordance with NAVFAC DM-3.10. Current NAVFAC criteria limit the maximum hangar bay noise level to 65 dBA. Noise levels of 55 dBA in the hangar bay with the ventilation system operating at maximum airflow have been achieved with careful design attention to fan and duct noise characteristics. Outdoor noise levels should not exceed 75 dBA and shall comply with local regulations.

3.5 Electrical Requirements. Electrical systems shall be provided in accordance with the NAVFAC DM-4 Series and as defined in paras. 3.5.1 through 3.5.3.

3.5.1 Electrical Installations. All electrical installations in the following areas shall meet the requirements in NFPA-70 for the specific hazardous (classified) location:

a) Hangar area.

b) Paint and chemical mixing rooms.

c) Paint equipment cleaning room.

d) Paint storage room.

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

3.5.2 Lighting. The electrical requirements for lighting shall be as follows:

a) Interior lighting in the main hangar bays shall provide 100 foot-candles measured 30 in. (0.76m) above the floor, with lighting intensities elsewhere in accordance with MIL-HDBK-1190. Use of metal halide lights in the hangar bays for proper color rendition is recommended. Future aircraft finishes may require other types of lights.

b) Exterior lighting shall be high-pressure sodium vapor where practical and shall be in accordance with the NAVFAC DM-4 Series.

c) Provide power outlets in the hangar bays for task lights such as underwing lighting.

3.5.3 Grounding. Two flush floor-ground receptacles, each with a 3/4-in. (19-mm) diameter ground rod, 10 ft (3.05 m) minimum length, shall be provided near the center of the aircraft position, with one on each side of the positioned aircraft. Resistance to ground shall be 25 ohms maximum. Bond all ground receptacles together with No. 4 AWG bare copper in or below the hangar floor.

3.6 Weight-Handling Equipment. The monorail hoist shall be in accordance with criteria in NAVFAC DM-38.01 and shall be of sparkproof construction with an explosion-proof motor. Controls shall be operable from the floor level and, when electric, shall meet the requirements indicated in para. 3.5.1.

3.7 Fire Protection. The fire protection design shall be in accordance with NAVFAC MIL-HDBK-1008, Fire Protection for Facilities Engineering Design and Construction. The following features shall be provided:

Hangar Bays (OH Spaces):

a) Protect the areas using aqueous film-forming foam (AFFF) deluge sprinkler systems or pre-action AFFF sprinkler systems using standard water sprinkler heads. The discharge rate shall be a minimum of 0.16 gpm/ft^2 (0.010 L/s per 0.0929 m^2) of air-foam solution.

b) Where pre-action AFFF sprinkler systems are used, water supply in addition to hose streams and monitor nozzles shall be sufficient to satisfy volume of water required by the number of sprinkler heads opened in accordance with one of the following:

(1) Within an 8,000 ft^2 (743 m^2) area where roof or ceiling height is 25 ft (7.6 m) or less above the floor level.

(2) Within an 18,000 ft^2 (1,672 m^2) area where roof or ceiling height is over 25 ft (7.6 m) but not more than 75 ft (22.8 m) above the floor level.

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the current version before use.

(3) Within a 35,000 ft^2 (3,252 m^2) area where roof or ceiling height is over 75 ft (22.8 m) above the floor level.

c) Supplementary low-level, fixed, AFFF nozzle systems shall be provided in main hangar areas. Oscillating nozzles may be provided instead of fixed nozzles if the use of fixed nozzles is not feasible. These systems shall be designed at a minimum application rate of 0.10 gpm/ft^2 (0.006309 L/s per 0.0929 m^2). These systems shall be activated by the overhead detection system. Strategically located manual controls shall also be provided.

d) Foam concrete tanks, proportioning equipment, and deluge valves shall be separated from main hangar by construction having a minimum of one-hour fire resistance rating.

e) Automatic, wet-pipe sprinkler system shall be provided in areas of the hangar facility not requiring overhead AFFF sprinkler protection.

Provide manual and automatic fire alarm system reporting to the base-wide system.

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the current version before use.

APPENDIX A
FACILITY PLATES

Cat. Code	Sheet Number	Title	Supersedes DD	Page
Section 1: Introduction				
211-54	1 of 4	Aviation Armament Shop	1291741	61
211-54	2 of 4	Aviation Armament Shop	1291741	62
211-54	3 of 4	Aviation Armament Shop	1291741	63
211-54	4 of 4	Aviation Armament Shop	1291741	64
211-54	1 of 5	Aviation Armament Shop	1291742	65
211-54	2 of 5	Aviation Armament Shop	1291742	66
211-54	3 of 5	Aviation Armament Shop	1291742	67
211-54	4 of 5	Aviation Armament Shop	1291742	68
211-54	5 of 5	Aviation Armament Shop	1291742	69
211-54	1 of 1	Aviation Armament Shop - Small	N/A	70
211-54	1 of 1	Aviation Armament Shop - Medium	N/A	71
211-54	1 of 1	Aviation Armament Shop - Large	N/A	72
Section 2: Maintenance Hangar				
Section 3: Corrosion Control Hangar				
211-03	1 of 4	Corrosion Control Hangar Type "A" Small	1403808	72a
211-03	2 of 4	Corrosion Control Hangar Type "A" Small	1403808	72b
211-03	3 of 4	Corrosion Control Hangar Type "A" Small	1403808	72c
211-03	4 of 4	Corrosion Control Hangar Type "A" Small	1403808	72d
211-03	1 of 4	Corrosion Control Hangar Type "B" Large	1403809	72e
211-03	2 of 4	Corrosion Control Hangar Type "B" Large	1403809	72f
211-03	3 of 4	Corrosion Control Hangar Type "B" Large	1403809	72g
211-03	4 of 4	Corrosion Control Hangar Type "B" Large	1403809	72h
Section 5: Paint Finishing Hangar				
211-04	1 of 4	Pre-Engineered Maintenance Hangar	1403097	73
211-04	2 of 4	Pre-Engineered Maintenance Hangar	1403097	74
211-04	3 of 4	Pre-Engineered Maintenance Hangar	1403097	75
211-04	4 of 4	Pre-Engineered Maintenance Hangar	1403097	76
211-04	1 of 4	Pre-Engineered Maintenance Hangar	1403098	77
211-04	2 of 4	Pre-Engineered Maintenance Hangar	1403098	78
211-04	3 of 4	Pre-Engineered Maintenance Hangar	1403098	79
211-04	4 of 4	Pre-Engineered Maintenance Hangar	1403098	80
211-04	1 of 4	Pre-Engineered Maintenance Hangar	1403099	81
211-04	2 of 4	Pre-Engineered Maintenance Hangar	1403099	82
211-04	3 of 4	Pre-Engineered Maintenance Hangar	1403099	83
211-04	4 of 4	Pre-Engineered Maintenance Hangar	1403099	84

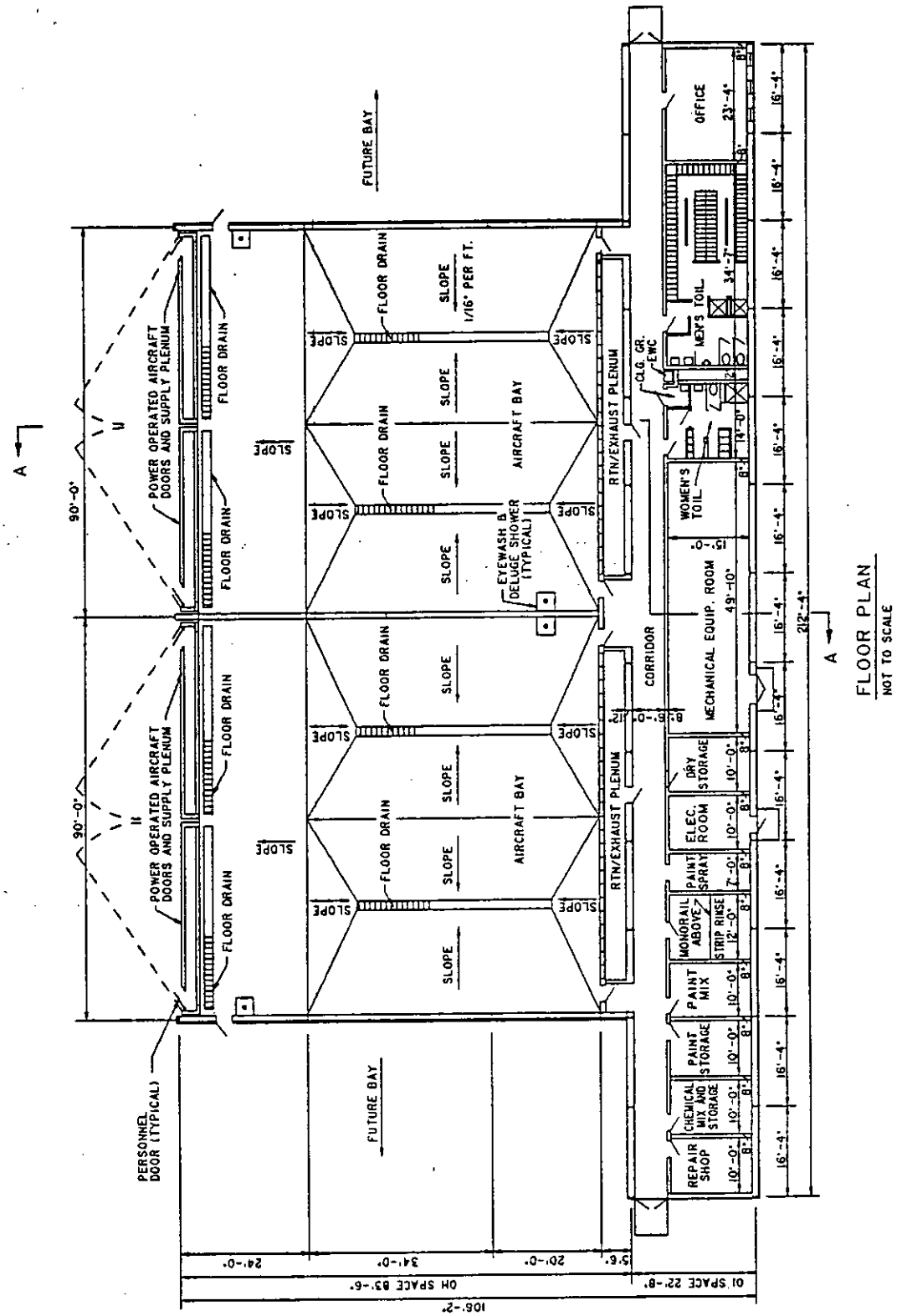
Source: https://assist.dla.mil -- Downloaded: 202610-26T1[illegible]
Check the source to verify that this is the current version before use.

APPENDIX A (Continued)

Cat. Code	Sheet Number	Title	Supercedes DD	Page
Section 4: Aircraft Weapons Alignment Facility				
211-82	1 of 5	Aircraft Weapons Alignment Shelter	1291733	85
211-82	2 of 5	Aircraft Weapons Alignment Shelter	1291733	86
211-82	3 of 5	Aircraft Weapons Alignment Shelter	1291733	87
211-82	4 of 5	Aircraft Weapons Alignment Shelter	1291733	88
211-82	5 of 5	Aircraft Weapons Alignment Shelter	1291733	89
211-82	1 of 6	Aircraft Weapons Alignment Shelter	1291734	90
211-82	2 of 6	Aircraft Weapons Alignment Shelter	1291734	91
211-82	3 of 6	Aircraft Weapons Alignment Shelter	1291734	92
211-82	4 of 6	Aircraft Weapons Alignment Shelter	1291734	93
211-82	5 of 6	Aircraft Weapons Alignment Shelter	1291734	94
211-82	6 of 6	Aircraft Weapons Alignment Shelter	1291734	95

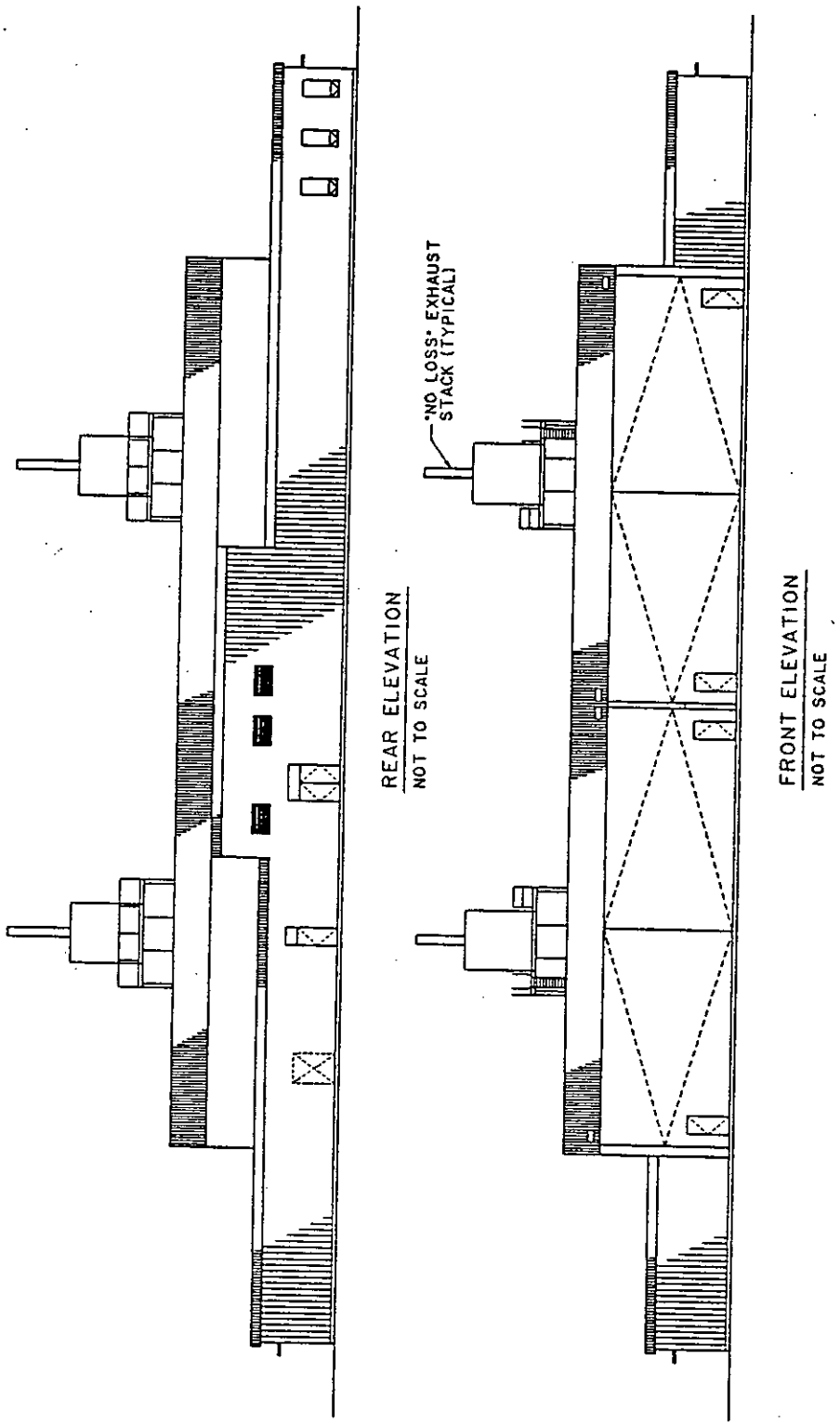
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

REPRINTED WITHOUT CHANGE



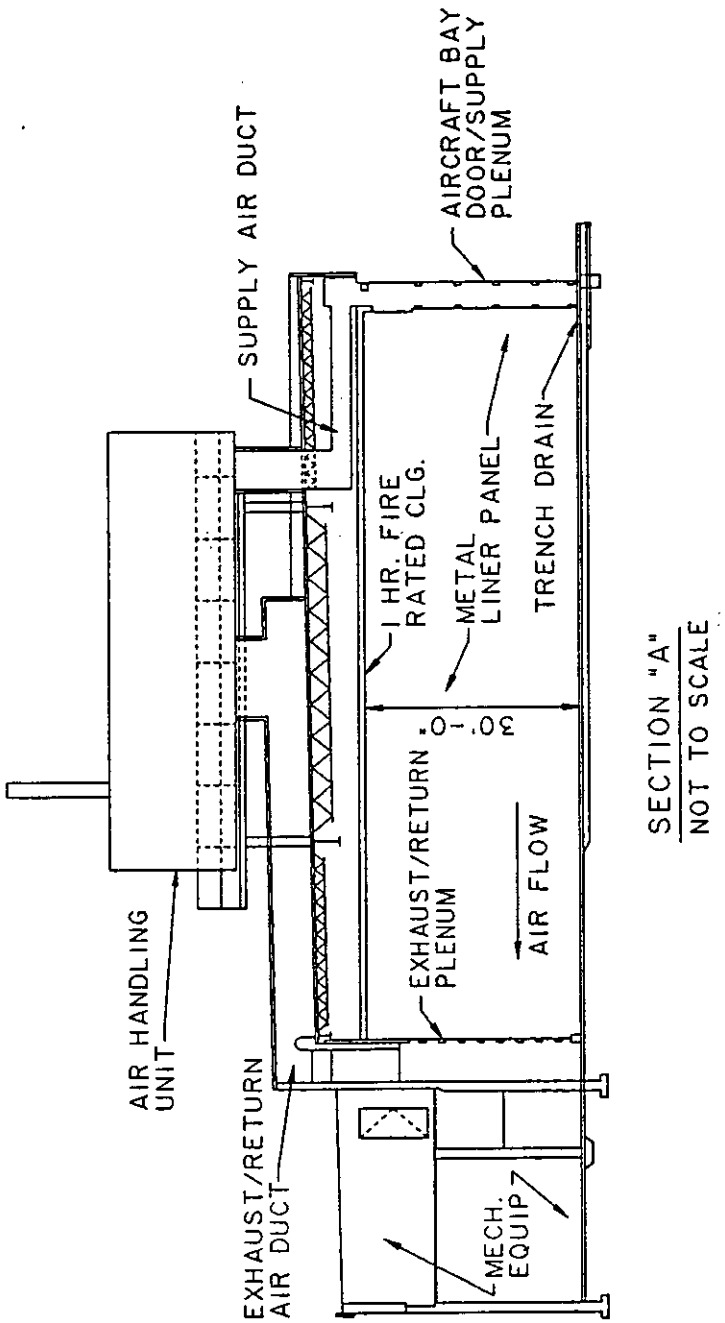
TITLE	DATE	FACILITY PLATE NO.	SHEET
Corrosion Control Hangar Type "A" Small		211-03	1 Of 4

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the current version before use.



TITLE	DATE	FACILITY PLATE NO.	SHEET
Corrosion Control Hangar Type "A" Small		211-03	2 Of 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.



TITLE	DATE	FACILITY PLATE NO.	SHEET
Corrosion Control Hangar Type "A" Small		211-03	3 Of 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

NOTES

PLUMBING REQUIREMENTS

COLD WATER	65 G.P.M.
HOT WATER (01 SPACE)	40 G.P.H.
RECOVERY RATE (THRU 100°F RISE)	
STORAGE	40 GAL.

FIRE PROTECTION REQUIREMENTS

SPRINKLERS (FOAM-WATER)	1123 G.P.M.
OSCILATING NOZZLES	702 G.P.M.
HOSE STREAMS	500 G.P.M.
TOTAL (EACH BAY)	2325 G.P.M.

HEATING REQUIREMENTS *

INSIDE DESIGN TEMPERATURE	
OH SPACE	85°F
01 SPACE	68°F
OUTSIDE DESIGN TEMPERATURE	22°F
HEATING LOAD	
OH SPACE	36,740,000 BTU/HR (W/O HEAT RECOVERY)
01 SPACE	167,000 BTU/HR

AIR CONDITIONING REQUIREMENTS *

INSIDE DESIGN TEMPERATURE	76°F.D.B.
INSIDE DESIGN HUMIDITY	50%
OUTSIDE DESIGN TEMPERATURE	91°F.D.B.
	76°F.W.B.
COOLING LOAD, OFFICE	12,000 BTU/HR

ELECTRICAL REQUIREMENTS (KW)

	MODULE OH SPACE	01 SPACE
LIGHTS		
CONNECTED LOAD	40	4
POWER		
CONNECTED LOAD	1289	4
TOTAL		
CONNECTED LOAD	1329	8
ESTIMATED DEMAND	1273	4
ADDITIONAL DEMAND FOR AIR CONDITIONING, OFFICE		2

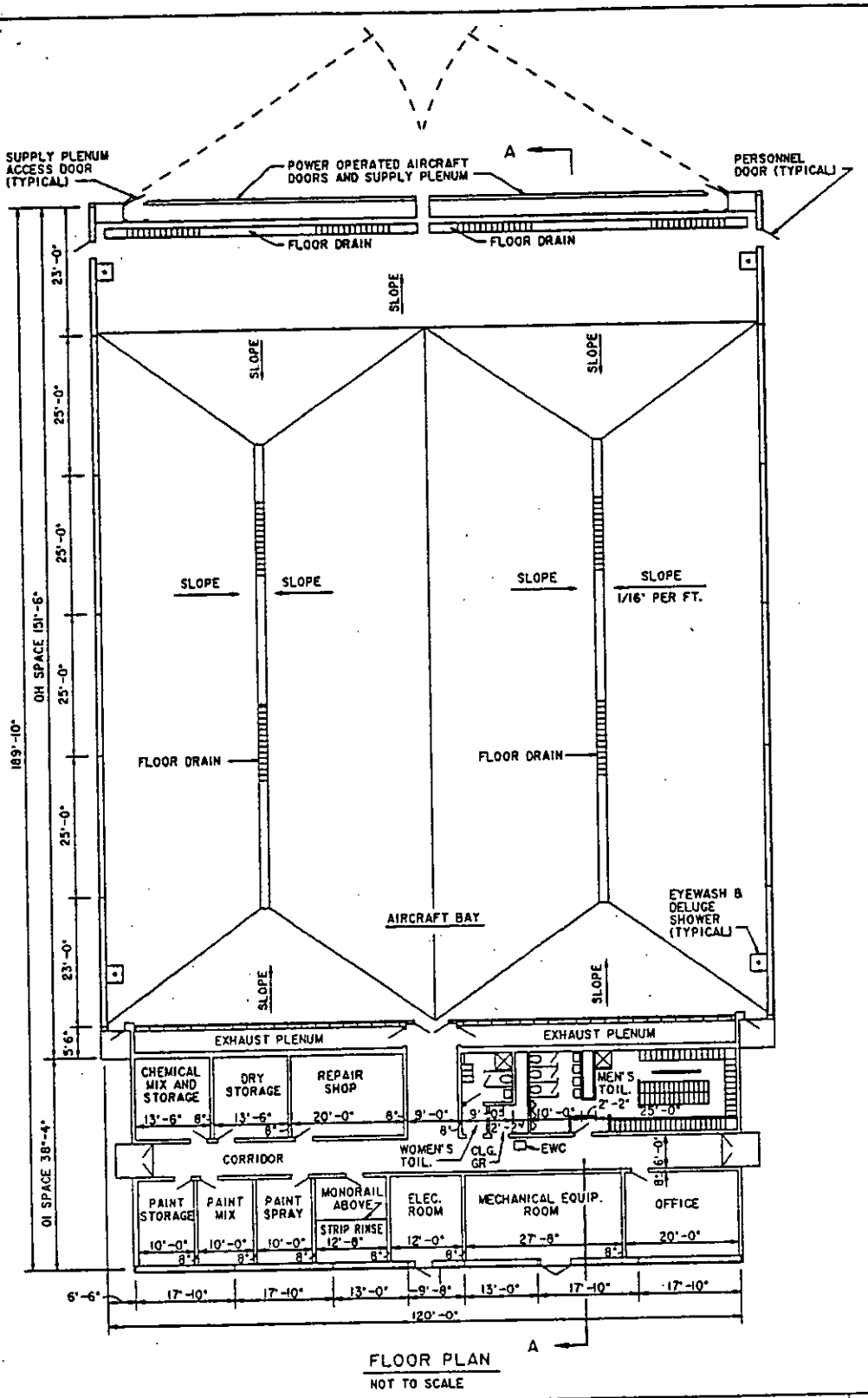
BAY DIMENSIONS

BAY DIMENSIONS SHOWN WILL SUPPORT S-3 AIRCRAFT, AND PROVIDE 10 FOOT HORIZONTAL AND 6 FOOT VERTICAL CLEARANCE BETWEEN AIRCRAFT AND HANGAR BAY WALLS/DOORS/CEILINGS. REDUCED CLEARANCES SHALL NOT BE USED WITHOUT PRIOR NAVFAC APPROVAL.

* REQUIREMENTS BASED ON NORFOLK, VIRGINIA AREA. PROVIDE COOLING FOR OFFICE SPACE ONLY. OH SPACE HEATING REQUIREMENTS VARY. VERIFY SPECIFIC FACILITY REQUIREMENTS.

TITLE	DATE	FACILITY PLATE NO.	SHEET
Corrosion Control Hangar Type "A" Small		211-03	4 Of 4

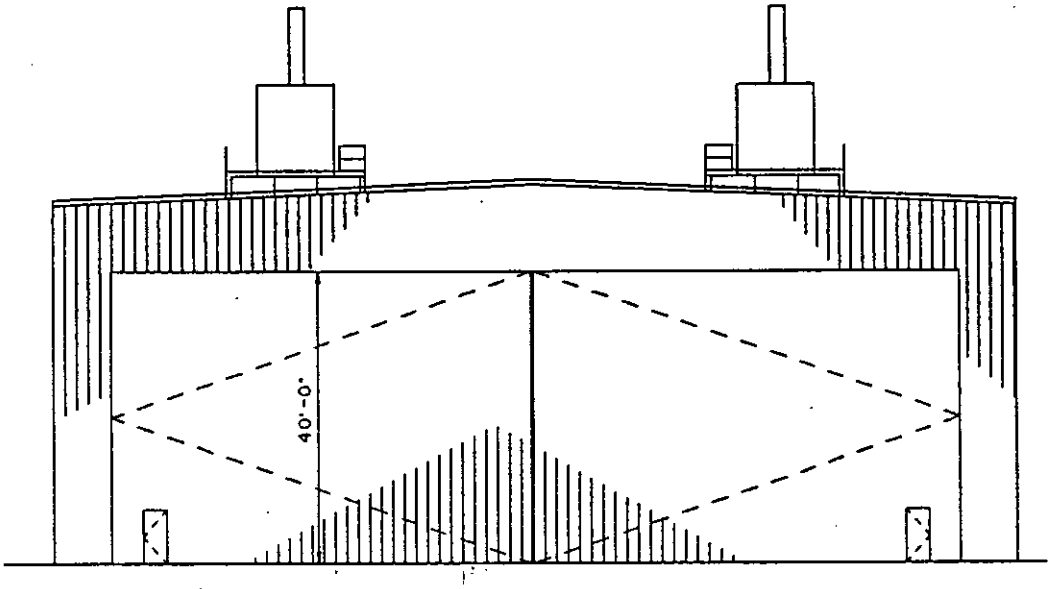
Source: https://assist.dla.mil - Downloaded: 2026-10-26T13:59Z
Check the source to verify that this is the current version before use.



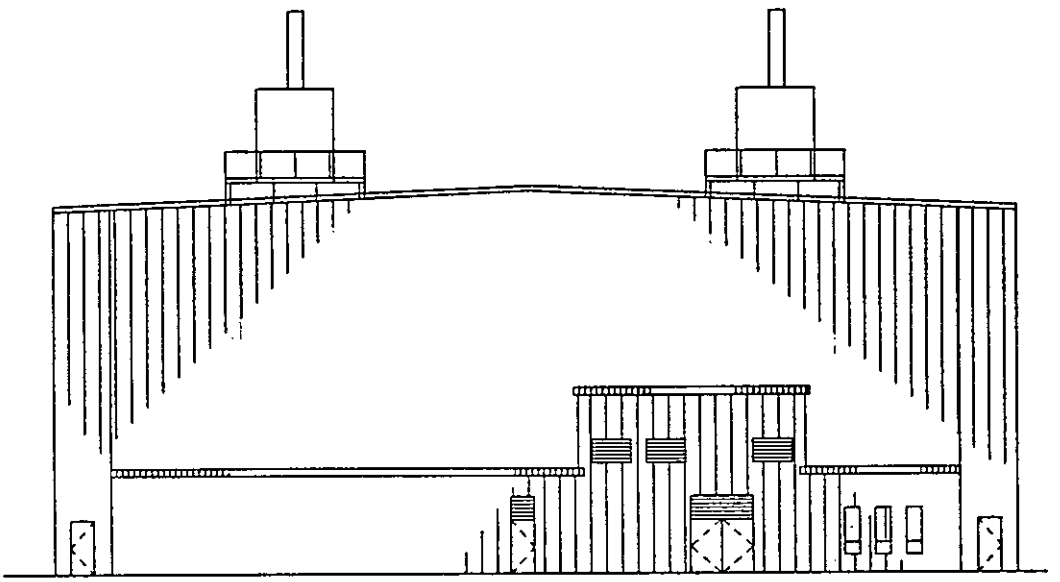
FLOOR PLAN
NOT TO SCALE

TITLE	DATE	FACILITY PLATE NO.	SHEET
Corrosion Control Hangar Type "B" Large		211-03	1 Of 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.



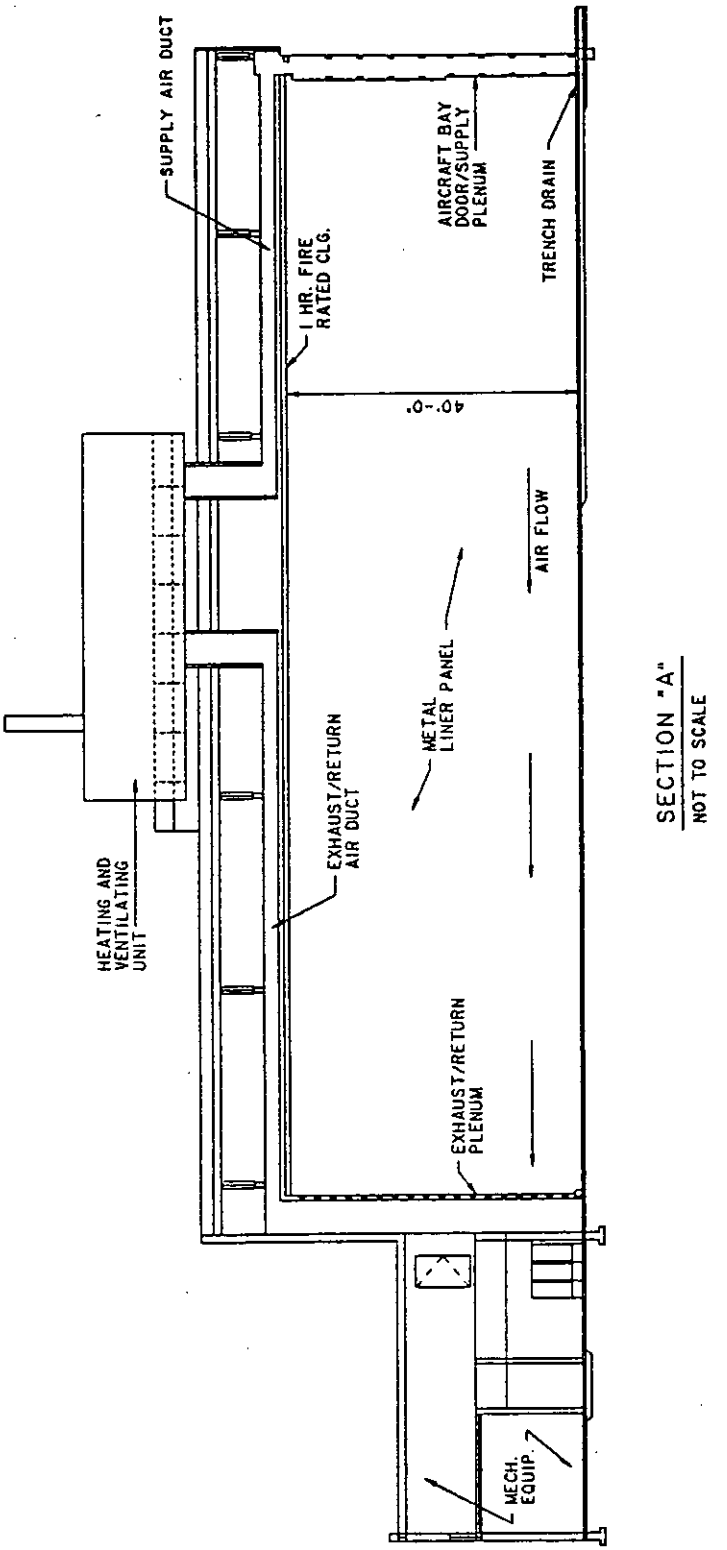
FRONT ELEVATION
NOT TO SCALE



REAR ELEVATION
NOT TO SCALE

TITLE	DATE	FACILITY PLATE NO.	SHEET
Corrosion Control Hangar Type "B" Large		211-03	2 Of 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.



TITLE	DATE	FACILITY PLATE NO.	SHEET
Corrosion Control Hangar Type "B" Large		211-03	3 Of 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

NOTES

PLUMBING REQUIREMENTS

COLD WATER	80 G.P.M.
HOT WATER (01 SPACE)	55 G.P.H.
RECOVERY RATE (THRU 100° F RISE)	
STORAGE	55 GAL.

FIRE PROTECTION REQUIREMENTS

SPRINKLERS (FOAM-WATER)	2803 G.P.M.
OSCILATING NOZZLES	1752 G.P.M.
HOSE STREAMS	500 G.P.M.
TOTAL (EACH BAY)	4555 G.P.M.

HEATING REQUIREMENTS *

INSIDE DESIGN TEMPERATURE	
OH SPACE	85°F
01 SPACE	68°F
OUTSIDE DESIGN TEMPERATURE	22°F
HEATING LOAD	
OH SPACE	32,660,000 BTU/HR (W/O HEAT RECOVERY)
01 SPACE	150,000 BTU/HR

AIR CONDITIONING REQUIREMENTS *

INSIDE DESIGN TEMPERATURE	76°F.D.B.
INSIDE DESIGN HUMIDITY	50%
OUTSIDE DESIGN TEMPERATURE	91°F.D.B.
	76°F.W.B.
COOLING LOAD, OFFICE	12,000° BTU/HR

ELECTRICAL REQUIREMENTS (KW)

	MODULE OH SPACE	MODULE 01 SPACE
LIGHTS		
CONNECTED LOAD	55	4
POWER		
CONNECTED LOAD	898	4
TOTAL		
CONNECTED LOAD	953	8
ESTIMATED DEMAND	877	4
ADDITIONAL DEMAND FOR AIR CONDITIONING OFFICE		2

BAY DIMENSIONS

BAY DIMENSIONS SHOWN WILL SUPPORT P-3 AIRCRAFT, AND PROVIDE 10 FOOT HORIZONTAL AND 6 FOOT VERTICAL CLEARANCE BETWEEN AIRCRAFT AND HANGAR BAY WALLS/DOORS/ CEILINGS. REDUCED CLEARANCES SHALL NOT BE USED WITHOUT PRIOR NAVFAC APPROVAL.

* REQUIREMENTS BASED ON NORFOLK, VIRGINIA AREA. PROVIDE COOLING FOR OFFICE SPACE ONLY. OH SPACE BAY HEATING REQUIREMENTS VARY, VERIFY SPECIFIC FACILITY REQUIREMENTS.

TITLE	DATE	FACILITY PLATE NO.	SHEET
Corrosion Control Hangar Type "B" Large		211-03	4 Of 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

REFERENCES

American National Standards Institute, Inc. (ANSI), 1430 Broadway, New York, NY 10018.

Z358.01-81 American National Standard for Emergency Eyewash and Shower Equipment

American Society of Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE), 1791 Tullie Circle NE., Atlanta, GA 30329.

Standard 52-76 Method for Testing Air Cleaning Devices Used in General Ventilation for Removing Particulate Matter

American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103.

D523-80 Standard Test Method for Specular Glass

Approval Guide, Factory Mutual Engineering Corporation, 1151 Boston-Providence Turnpike, P.O. Box 688, Norwood, MA 02062.

Department of Defense (DOD), Copies can be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

DOD-STD-1399 Interface Standard for Shipboard Systems, Electric Power, Alternating Current

Department of the Navy

Chief of Naval Operations (OPNAV)

OPNAVINST 4790.2D The Naval Aviation Maintenance Program

OPNAVINST 5510.1G Department of the Navy Information Security Program Regulation

OPNAVINST 5530.14 Physical Security and Loss Protection

Naval Air (NAVAIR) Systems Command Technical Manuals. Available from the Department of the Navy, Naval Air Systems Command, Washington, DC 20360.

01-1A-22 Organizational, Intermediate, and Depot Maintenance, Aircraft Radomes and Antenna Covers

01-1A-509 Aircraft Weapons Systems Cleaning and Corrosion Control

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the currentversion before use.

01-1A-512	Design Guide for Avionics Shop Power Distribution
13-1-6.2	Manual Aviation - Crew Systems Parachutes
16-1-529 Volume 1	Electromagnetic Radiation Hazards (Hazards to Personnel, Fuel, and Other Flammable Material)

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the currentversion before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:56Z
Check the source to verify that this is the current version before use.

REFERENCES (Continued)

Naval Facilities Engineering Command (NAVFACENGCOM) Design Manuals (DM), Naval Facilities Guide Specification (NFGS), Publications (P), Military Handbooks (MIL-HDBK), Naval Facilities Instructions (NAVFACINST), Naval Sea Systems Command (NAVSEA), and Naval Supply Systems Command (NAVSUP).

DM-1 Series	Architecture
DM-1.01	Basic Architectural Requirements and Design Considerations
MIL-HDBK-1001/2	Materials and Building Components
DM-1.03	Architectural Acoustics
DM-2 Series	Structural Engineering
DM-3.01	Plumbing Systems
DM-3.03	Heating, Ventilating, Air Conditioning, and Dehumidifying Systems
DM-3.05	Compressed Air and Vacuum Systems
DM-3.10	Noise and Vibration Control of Mechanical Equipment
DM-4 Series	Electrical Engineering
DM-5.08	Civil Engineering - Pavements
DM-5.09	Civil Engineering Metering, Instrumentation and Control, and Chemical Feeding
DM-5.12	Civil Engineering - Fencing, Gates, and Guard Towers
MIL-HDBK-1008	Fire Protection for Facilities Engineering Design and Construction
DM-21 Series	Airfield Pavement Design
DM-22	Petroleum Fuel Facilities
DM-23 Series	Airfield Lighting and Navigational Aids
DM-28.04	General Maintenance Facilities
DM-28.05	Environmental Control - Design of Clean Rooms

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:56Z
Check the source to verify that this is the current version before use.